Event-Level Predictive Planning for Continuous Driving and Gap Crossing of a Single-Track Robot

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Abstract— Traversing a gap with a single-track robot requires more than reaching the opposite bank: the landing state must also permit subsequent driving. We present an event-level predictive planning framework that composes a frozen ground controller and a frozen jumping skill on a known off-road map. A compact residual world model predicts the occurrence, timing, and resulting state of controller entry, flight, far-bank contact, handoff, and goal continuation. The planner evaluates route, approach-speed, preference, and skill candidates through sampled event rollouts, while accounting for vehicle motion during computation. Supervised training uses physical executions with starting-condition-grouped splits. On 240 stored executions from four held-out starting conditions, the model obtains a far-contact position error of 0.104 m and an autoregressive completion Brier score of 0.0957, versus 0.1216 for a training-rate constant. Online planning completes five of twelve tasks, compared with six for a training-derived distance rule. These results establish an executable prediction–decision–handoff pipeline and expose its failure modes; they do not establish a completion-rate advantage. The framework provides a tractable interface for studying how intermediate contact outcomes constrain the composition of existing locomotion skills.

Index terms—Model learning for control; motion planning; hybrid locomotion; single-track robots.

I. Introduction

A single-track robot approaching a gap must coordinate steering, lean, and speed before leaving the ground. A short-lived crossing is not sufficient for a traversal task: after contact with the opposite bank, the robot must resume driving within the available road corridor. The upstream decision therefore concerns the state delivered to a jumping skill and the downstream consequences of that skill. A path that is geometrically connected can still be dynamically difficult to enter, and a successful landing can still be followed by task failure.

Controller composition offers a way to reuse competent motion skills instead of learning a monolithic policy for every route. Sequential composition has long provided a framework for organizing robot behaviors [1]. For off-road single-track traversal, the interface between skills includes a transient physical process: preparation, flight, contact, and continued balancing. Evaluating only the final success label hides where this process fails. Conversely, predicting every simulator step may be unnecessary when planning only selects among a finite set of routes and skill combinations.

We investigate whether a compact event-level world model can connect these two scales. Our model receives simulator state, causal controller history, local terrain, and a candidate task action. It predicts milestone occurrence, time, and post-event state. These outputs are propagated through a prescribed event sequence to rank candidates before actual execution. Low-level controllers remain frozen; the model neither generates motor commands nor synthesizes a new path. We use "world model" in the restricted sense of an action-conditioned predictor of physical outcomes, not a visual generative model or universal simulator.

Our contribution is an implemented predictive composition interface for continuous driving and gap crossing: (i) a residual event-conditioned model trained from successful and failed physical executions; (ii) finite candidate evaluation with computation-delay propagation and controller handoff; and (iii) a simulation study relating intermediate predictions to complete tasks on a heterogeneous map. We evaluate prediction and execution separately. The planner executes autonomous selections and completes both crossing and bypass tasks, but it does not outperform a distance rule in the reported online panel.

II. Related Work

Composing locomotion behaviors. Sequential behavior composition organizes dynamical controllers into larger tasks [1]. For wheeled bipeds, Xin and Vijayakumar combine rolling and contact-changing template models with online planning and whole-body control [2]. Lee et al. integrate learned locomotion and navigation for wheeled-legged robots [3]. Our setting differs in morphology and interface: a single-track vehicle uses an existing ground controller and fixed jumping policy, while the upper layer evaluates their predicted event-level consequences. These precedents motivate composition, rather than establishing a claim to first introducing it.

Learned models for planning. PETS combines probabilistic dynamics with sampled uncertainty propagation [4]. Yang et al. study learned locomotion models and account for planning latency [5]. DayDreamer learns world models for physical robots, including locomotion and navigation [6]. Here, learning is supervised on simulator executions, and the action is a route–skill decision rather than a low-level control sequence. The model predicts selected milestones and does not train an imagined policy. A direct success predictor is retained as a comparator: intermediate predictions should be evaluated beyond their ability to assign a final score.

III. Task and System Interface

A. Common-goal traversal

Let $\mathcal{H}$ be a known height map with support mask, $\mathcal{R}$ a finite road set, and $\mathcal{G}$ a shared goal. A task action is

$$a = (r, v^\star, \alpha, \kappa), \quad r \in \mathcal{R}, \quad \kappa \in \{0, 1\}, \tag{1}$$

where $v^\star$ and α specify approach speed and ground-controller preference, and κ selects ground-only traversal or a ground–jump–ground sequence. Every route targets the same region. The robot continues moving during planning; stopping or restoring a saved physical state is not used to resolve a late decision.

Task success requires reaching $\mathcal{G}$ at the declared minimum forward speed, without physical termination or any recorded corridor violation:

$$Y = \mathbf{1}_{\mathcal{G}} \mathbf{1}_{\text{physical valid}} \mathbf{1}_{\text{corridor valid}}. \tag{2}$$

Far-bank contact is intermediate, not success. Bypasses need not generate flight or landing events. Failure endpoints and milestones are retained, so surviving briefly after contact is not conflated with task completion.

B. Frozen controllers and handoff

Ground motion uses the existing STTW H75 controller with ECBC/ESO and zero lower learned residual. Jumping uses a frozen heading-conditioned checkpoint, heading125. H75 adjusts speed/steering references from causal history and path preview above the balance controller and extended-state observer (ESO). Increasing α shifts the trained preference from path following toward speed tracking, with finite secondary costs and unchanged actuator limits. These identities are held fixed in the main study. At the jump-entry line, control switches to the jump policy; this line is distinct from observed takeoff. After far-bank contact, ground control resumes at the next 20-ms boundary. No physical state reset is performed. Controller initialization follows the existing handoff interface rather than assuming interchangeable ground and airborne histories.

The chosen $(v^\star, \alpha)$ controls approach. Postlanding motion uses fixed 1.5 m/s and $\alpha = 0.5$, not separately optimized settings. The planning objective thus concerns the consequences of approach decisions for later continuation. The continuation predictor targets the final goal in (2); it is not a dedicated estimator of stabilization distance.

IV. Event-Level World Model

A. State, terrain, and candidate encoding

The encoded state is

$$s = [p^\top, q^\top, v^\top, \omega^\top, z^\top, b]^\top \in \mathbb{R}^{30}, \tag{3}$$

where (p, q, v, ω) is the 13-dimensional root state, q an $xyzw$ quaternion, $z \in \mathbb{R}^{16}$ a history summary, and b prior corridor violation. The 80-dimensional observation and 352-dimensional causal controller history are standardized with TRAIN statistics and projected by TRAIN-only PCA. History includes ESO variables, previous references and commands, policy filters, and controller frames. Compression reduces model size but does not establish a sufficient Markov state.

TABLE I

Supervised transitions. Ground-only actions use index 4 directly.

m	Input milestone	Target milestone
0	Approach state	Jump-controller entry
1	Controller entry	Confirmed flight
2	Flight	First far-bank contact
3	Far-bank contact	STTW handoff
4	Handoff / ground state	Successful goal continuation
5	Controller entry	Clearance (auxiliary)

Terrain input is an 8×8 yaw-aligned crop over forward offsets $[-0.2, 1.8]$m and lateral offsets $[-0.5, 0.5]$m. Triangle interpolation samples the collision mesh. The 128-vector contains heights in millimeters and a support mask; zero height with zero support denotes missing ground. It is known-map information, not a learned perception output.

The predictor represents each route by five summaries: maximum and minimum lateral position, longitudinal position of maximum absolute lateral displacement, and first and last nonzero lateral positions. These, $v^\star$, α, and κ form eight features. Distances to the near lip, far lip, and goal augment them to $\xi(s, a) \in \mathbb{R}^{11}$. The controller follows the full route; only the predictor uses compressed geometry.

B. Events and model outputs

A supplied event-transition index $m \in \{0, \ldots, 5\}$ specifies Table I. Ordering is prescribed. The model does not infer arbitrary next contact modes or learn a categorical transition graph. For a transition, the network returns

$$(\mu_\Delta, \ell_\Delta, \ell_\tau, \ell_h) = F_\theta(s, \xi, H, m), \tag{4}$$

namely 30 residual means, 30 residual log variances, one transformed event time, and one occurrence logit. Occurrence is estimated by $\hat{h} = \sigma(\ell_h)$; time by

$$\hat{\tau} = \exp[\text{clip}(\ell_\tau, 0, \log 11)] - 1. \tag{5}$$

Time is a deterministic estimate in $[0, 10]$s. Residual uncertainty is diagonal Gaussian; no ensemble or calibrated confidence set is assumed.

State prediction combines a physical baseline with a learned residual:

$$\hat{s}^+ = f_0(s, \hat{\tau}, m) + \bar{d} + \sigma_d \odot \delta, \tag{6}$$

$$\delta \sim \mathcal{N}(\mu_\Delta, \text{diag}(e^{\ell_\Delta})). \tag{7}$$

The baseline advances position with current velocity and holds other entries. Ground indices 0 and 4 hold vertical position. This is constant-velocity propagation, not a ballistic or full single-track model. The residual captures

departures, including contact-related state changes. Scaling $(\bar{d}, \sigma_d)$ is fitted on TRAIN; log variances are clipped to $[-4, 3]$. Sampled quaternions are normalized before the next step.

C. Architecture

The terrain encoder is $128 \to 32 \to 32$ with SiLU activations. Its output is concatenated with standardized state (30), candidate features (11), and event indicator (6). A shared $79 \to 128 \to 64$ MLP feeds three linear $64 \to 62$ heads. The approach head serves index 0, the obstacle head serves 1,2,3,5, and continuation serves 4. The network has 35,770 parameters. Shared features represent state–terrain context while separate heads map approach, contact-rich motion, and continuation. Specialization is tested against a single head, not presumed beneficial.

D. Training pairs and loss

Pairs are extracted from actual published-candidate executions. Confirmed flight requires an armed attempt, bilateral unloading (contact norm at most 1 N), clearance above 15 mm for 40 ms, and subsequent near-lip crossing with at least 5 cm advance. Far contact requires a valid attempt and filtered, vertically loaded far-bank patches (over 1 N) for at least one wheel. Clearance additionally requires both wheels to traverse the gap interior unloaded before passing the far lip. Thus flight is a confirmed-progress milestone rather than the first zero-contact sample. Let $(s_i, m_i, a_i, H_i, s_i^+, \tau_i, h_i)$ denote a transition, with $h_i = 1$ if the next milestone occurs. Physics-rate records supply flight, clearance, far-contact, and handoff labels; control history is stored at 50 Hz. Additional causal approach and ground-only prefixes occur every 25 control steps (0.5 s). A missing next event supplies a negative occurrence label. Its failure endpoint is stored, but not treated as a successful event-state sample.

For standardized residual target d_i relative to $f_0(s_i, \tau_i, m_i)$, the loss is

$$\mathcal{L}_x = \frac{1}{N_+} \sum_i \frac{h_i}{2D} \sum_{j=1}^{D} [(\mu_{ij} - d_{ij})^2 e^{-\ell_{ij}} + \ell_{ij}], \tag{8}$$

$$\mathcal{L}_\tau = \frac{1}{N_+} \sum_i h_i [\ell_{\tau,i} - \log(1 + \tau_i)]^2, \tag{9}$$

$$\mathcal{L}_h = \frac{1}{B} \sum_i \mathrm{BCEWithLogits}(\ell_{h,i}, h_i), \tag{10}$$

$$\mathcal{L} = \mathcal{L}_x + \mathcal{L}_\tau + \mathcal{L}_h, \tag{11}$$

where $D = 30$, B is batch size, and $N_+ = \max(1, \sum_i h_i)$. Positive events train state/time; all examples train occurrence. Residual scaling uses recorded TRAIN transitions, whereas conditional regression is positive-masked. Training conditions on the observed preceding milestone. This teacher-forced prediction differs from autoregressive use of sampled states in planning.

All prefixes and positions of a starting condition remain in one split. PCA uses TRAIN histories sampled at 10 Hz. Optimization uses Adam at 10^{-3}, batch 256 with replacement, gradient clipping 5, and 600 updates. DEV loss is evaluated every 100 updates; the lowest assessed checkpoint is selected. Main and shared-head models both select update 600. This is supervised fitting, not new PPO training. TEST data do not fit scaling, PCA, network parameters, or checkpoint selection.

V. Predictive Task Selection

A. Sampled event rollout

For each candidate, 32 particles start at the current state. Jump candidates propagate through indices 0–4; ground-only candidates use 4. Each step recomputes distances and terrain from predicted state, samples (7), and accumulates time. Common random numbers are shared across candidates. Predicted road history is clipped to $[0, 1]$ and made nondecreasing.

The implemented ranking estimate is

$$\widehat{Q}(s, a) = \frac{1}{P} \sum_{p=1}^{P} (1 - \hat{b}_{p,K}) \prod_{k \in \mathcal{K}(a)} \hat{h}_{p,k}. \tag{12}$$

It is a heuristic completion score, not a calibrated factorization: the goal label already includes corridor validity, so the final road factor overlaps that condition. Auxiliary clearance is logged but does not enter the product. The software also provides a time penalty; the online comparison uses (12), without time or energy penalties.

B. Admission and latency-aware publication

A candidate is admitted if the current position is within its road's 0.6-m half-width; jump candidates also require a position before the entry line. These are geometric checks, not dynamic feasibility guarantees. The vehicle may be close enough to a road to pass admission yet unable to follow its curvature.

Measured inference wall time t_c becomes $\max(1, \lceil t_c / \Delta t \rceil)$ control steps. The old controller executes those physical steps. At publication, admission is rechecked with the updated state, and the admissible candidate with the highest stored score is applied. Scores are not recomputed at publication. Rejected or closed-window decisions are recorded separately. Route publication reprojects the follower and updates approach speed/preference while preserving ongoing ESO and filter state. Trials contain one upper-level query per episode; continuous feedback occurs in the low-level controllers.

VI. Experimental Design

A. Map and physical execution

Isaac physics uses a 2.5-ms step and 20-ms control period. The 0.30-m gap spans $x = 10.3$–10.6m with equal-height banks. The jump zone $x = 8.5$–12m is

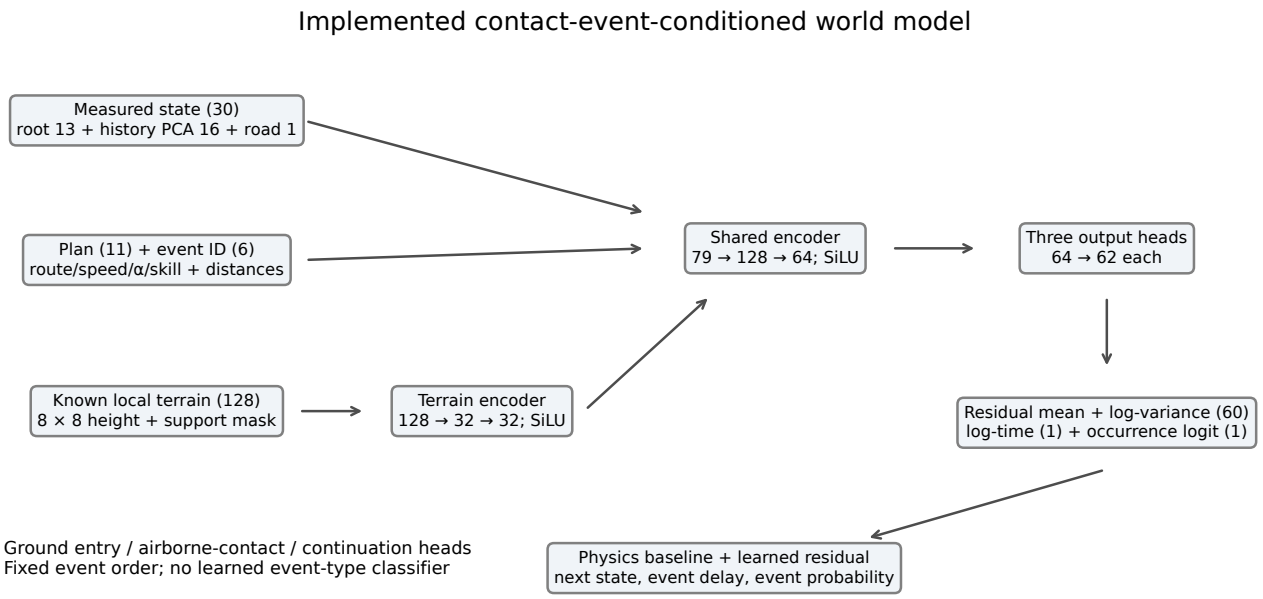


Fig. 1. Implemented model and execution interface. Event occurrence, time, and state outputs support candidate evaluation; frozen controllers produce motor commands. The event order is supplied, not discovered.

TABLE II

Executed planning procedure.

1. Observe root state, controller history, and known-map crop.
2. Encode state; enumerate route–speed–preference–skill actions.
3. Apply geometric admission; evaluate 32-particle event scores.
4. Continue old control for measured inference delay.
5. Recheck admission; publish highest stored-score admissible action, or retain current action.
6. Execute ground control; switch to jump at entry if selected.
7. At far contact, hand back to STTW; evaluate common goal and retain failure endpoints.

flat; approach and continuation have 1–4-mm collision stones and 2-cm/8-cm ridges. A side corridor permits continuous bypass. The goal is $x \in [16.5, 18.5]$m, $|y| \leq 0.5$m, with forward speed at least 0.5 m/s within the original 10-s horizon.

Straight crossing, S-bend crossing, rounded right-angle crossing, and bypass are combined with speeds $\{1.5, 1.8, 2.0, 2.3, 2.6\}$m/s and $\alpha \in \{0, 0.5, 1\}$:60 candidates. Query positions $x = 1.8, 3.0, 5.0$m change preparation and remaining time. The deadline is not restarted after publication. Physical termination includes body contact, nonfinite state, upright roll beyond 35°, pitch beyond 75°, root height below −0.15 m, or leaving the outer task bounds; ground-controller domain/numerical failures are also retained. Road departure beyond 0.6 m is latched and invalidates eventual task success.

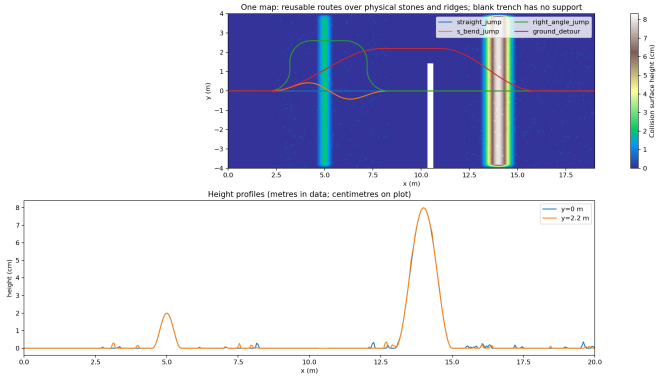


Fig. 2. Shared off-road task with crossing routes, bypass, and common goal. Stones and ridges are collision geometry outside the flat jump zone. Route existence does not imply dynamic executability.

B. Data roles and comparisons

Initial coverage comprises 12 starts×60 candidates (720 executions). Main collection comprises 8 starts×3 positions×60 requests (1,440 executions);6 starts are TRAIN,2 DEV. Of these,1,199 publish a candidate; only actual postpublication segments train the model. The other 241 remain rejected/closed-window coverage. Sixty finish under the original action and are not credited to an unpublished candidate.

For frozen prediction evaluation,240 stored executions from 4 other starting conditions are reused without fitting. These TEST roots were previously inspected for skill comparisons; they are outside model training but not previously unexamined confirmation data. Conditional event errors use observed previous events. Initial completion estimates use 32-particle autoregressive rollouts for the actually executed candidate. This is

prediction evaluation, not another online decision test.

The separate online panel freezes both predictors and a TRAIN distance rule, then evaluates 4 new starts at 3 query positions:12 trials per method but 4 independent starts. The direct comparator uses a $169 \to 64 \to 32 \to 1$ SiLU MLP on state, plan, and terrain, trained by binary cross entropy with Adam at 10^{-3} for 600 updates (batch 256, DEV assessment every 100). It reuses the frozen PCA but does not roll out events. The TRAIN rule selects bypass $(1.8, 0)$, S-bend crossing $(2.0, 0.5)$, and straight crossing $(1.8, 0.5)$ at the three positions, respectively; pairs denote speed in m/s and α. A 27,710-parameter shared-head ablation uses identical event inputs, labels, batch budget, and split; it changes specialization and capacity, not the event formulation.

Metrics are Brier score, conditional event-time and XY error, complete-task count, corridor violations, and query time. State/time error includes occurred events only; Brier includes missing events. Correlated prefixes do not increase independent-root count. The study is exploratory and single-map; no significance or equivalence claim is inferred from this small panel.

VII. Results

A. Physical task coverage

Initial collection completes 122/720 tasks:44/180 straight crossings,20/180 S-bends,58/180 bypasses, and 0/180 right-angle routes. These are complete physical task instances across several routes, not planner-selected success rates. The right-angle condition remains an explicit capability failure.

In multi-position collection, published candidates complete 162 tasks. Bypass publishes 120,120, and 0 times at early/middle/late positions, completing 41,7, and 0. Late exclusion is geometric, not a learned physical deadline. S-bend requests complete 12,16,15 tasks (120 requests per position). Differences include state, preparation, and remaining time, not a controlled distance-only effect.

B. Training and conditional event outputs

Fig. 3 shows saved training and DEV losses; only the six recorded checkpoints are plotted. Model outputs include event probabilities and predicted post-event states, not merely a final success score. Fig. 5 shows the event-time outputs; Fig. 4 compares predicted XY displacement against observed events on DEV. These are one-transition predictions from observed previous milestones; absent events contribute probability labels, not invented event-state targets.

On DEV, far-contact time MAE is 0.0349 s and XY error 0.1002 m; goal-continuation XY error is 0.8825 m. The larger continuation error motivates evaluating downstream predictions rather than treating landing as completion. All 121 applicable DEV far contacts hand off, so a near-certain constant outperforms the model's handoff-occurrence prediction. At far contact, conditional world-x velocity MAE is 0.318 m/s, with world-frame Euler roll/yaw MAE of 1.35°/1.56°. Goal-event values are 0.137 m/s and 3.72°/8.24°. Angles are derived from normalized quaternions with wrapped differences; world Euler roll includes the asset's nominal orientation and is not the controller's lean error.

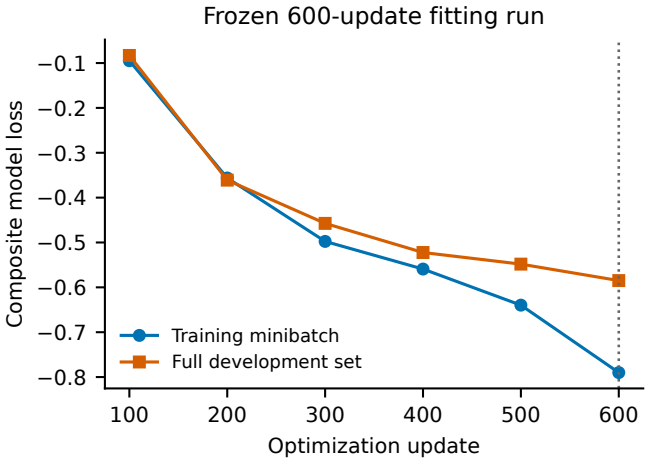


Fig. 3. Saved optimization measurements. DEV loss selects update 600. Curves contain six recorded checkpoints, not interpolated per-update evidence or proof of convergence.

TABLE III

Frozen prediction on four previously observed TEST roots. Three-head/shared-head errors; lower is better.

Target	Brier	Time MAE(s)	XY error(m)
Entry	.0480/.0482	.1776/.1448	.4829/.4915
Flight	.0174/.0162	.0675/.0805	.1404/.1956
Far contact	.0439/.0459	.0402/.0694	.1042/.1777
Handoff	.0382/.0352	.0342/.0238	.1520/.1793
Goal	.0888/.1027	.2912/.3517	.7839/1.8112
Clearance	.1848/.1914	.0704/.0902	.1960/.2478

C. Frozen held-out-root prediction

Table III evaluates both frozen models on 240 stored TEST executions with 34 completions. The branched model yields far-contact time error 0.0402 s and XY error 0.1042 m. Its initial autoregressive completion Brier is 0.09570, compared with 0.11455 for the shared-head model and 0.12160 for the TRAIN-rate constant. This separates out-of-training-root prediction evidence from DEV checkpoint selection. The branched model has lower XY error for all six events, but shared-head flight and handoff Brier are lower. Both are worse than the near-certain constant for handoff occurrence (79/79 events). Specialization is therefore not uniformly better across outputs. Parameter counts differ, and only one optimization seed is evaluated. The comparison supports descriptive prediction findings, not a capacity-controlled architecture advantage or new-map generalization.

D. Online decisions and continuation

The event planner chooses early bypasses and completes all 4. Late it selects straight crossings, one reaching the goal after far-bank handoff. For start 3, the early

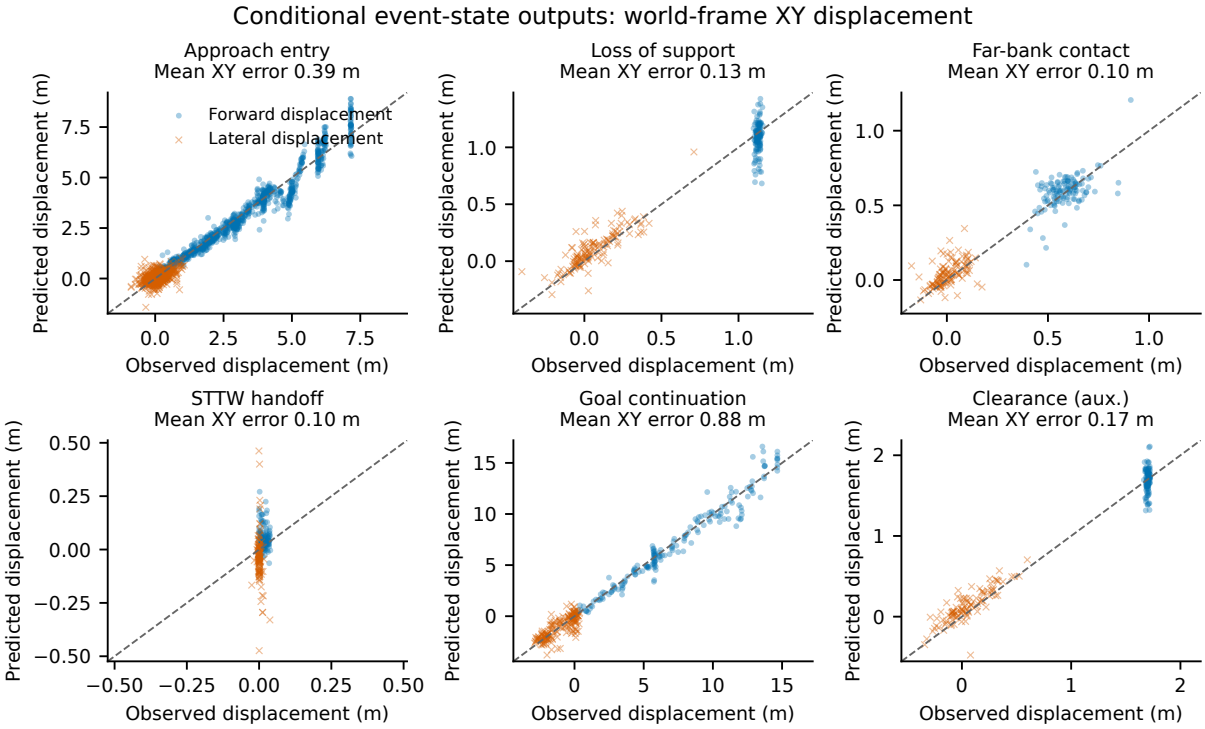


Fig. 4. Conditional world-model outputs versus observed event states on DEV. State targets exist only for occurred events. This evaluates individual transitions with observed preceding states; it is distinct from multi-event autoregressive forecasting. All declared events are retained.

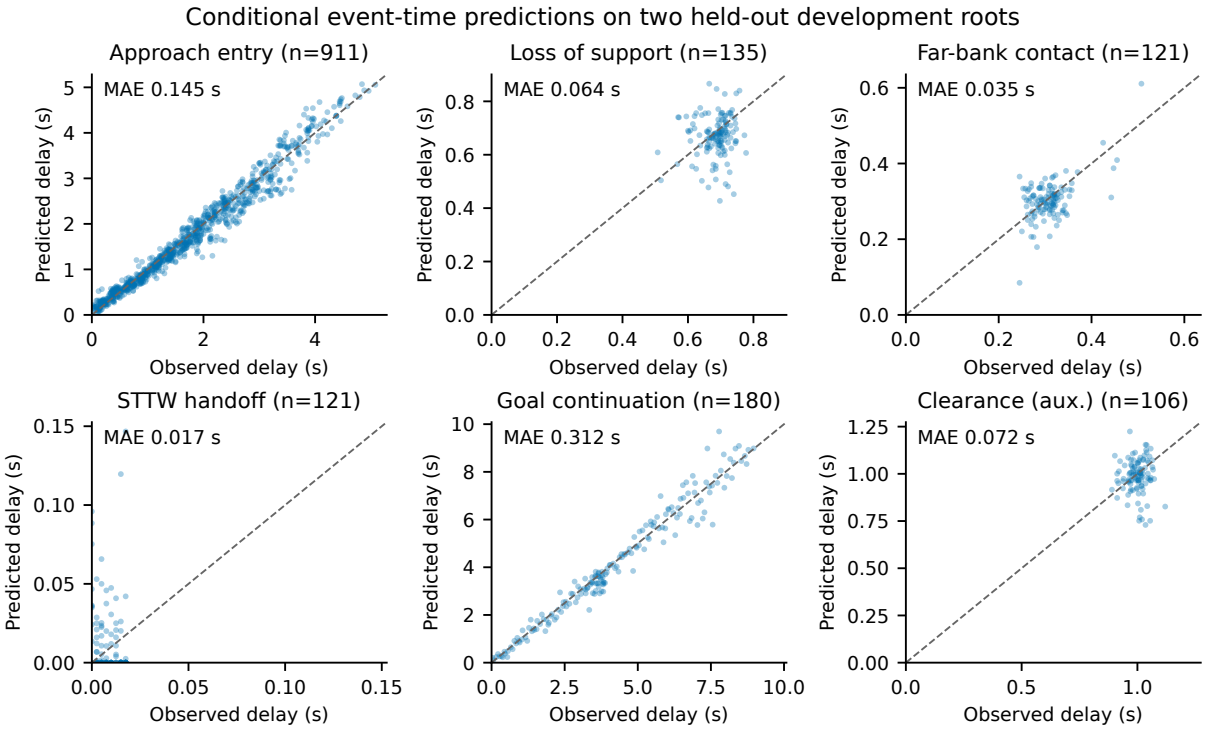


Fig. 5. Observed versus predicted time to each event on DEV, conditioned on the actual preceding state. Dashed lines indicate exact predictions. Failed next events train occurrence classification but have no event-time target; they are not assigned the terminal time in this display.

episode chooses bypass at 1.8 m/s,$\alpha = 0$; the late episode chooses straight crossing with the same approach pair. In the latter, jump entry occurs at 3.90 s and STTW handoff at 4.92 s before completion. These are distinct episodes sharing a starting-condition definition, not two choices in one rollout. The middle bypass for that start

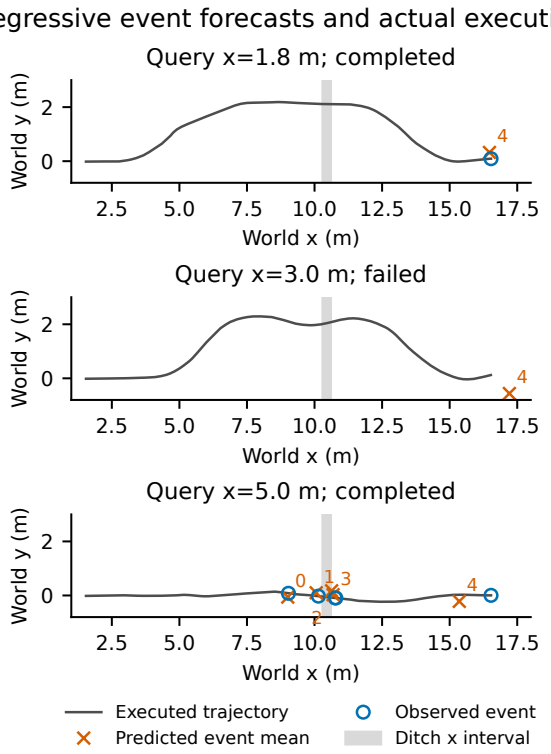


Fig. 6. Autoregressive world-model outputs and observed execution for start 3 at three decision positions. Early bypass and late crossing succeed; middle bypass fails. Candidate scores are 0.6874,0.3659,0.1971 respectively. Late bypass exclusion is geometric. These forecasts start from the query state, unlike the teacher-forced predictions in Fig. 4.

TABLE IV
Online complete-task outcomes: four new starting conditions repeated at each query position.

Query x (m)	Event model	Direct success	Distance rule	Event violations
1.8	4/4	3/4	4/4	0/4
3.0	0/4	0/4	1/4	4/4
5.0	1/4	2/4	1/4	2/4
Total	5/12	5/12	6/12	6/12

fails and remains in the aggregate.

Event and direct models each complete 5/12 tasks versus 6/12 for the distance rule, each gaining 1 and losing 2 relative to that rule. All 4 event-planned middle-position episodes violate the corridor. Thus geometric admission and model scores do not guarantee turn entry. Positive cases prove actual choice and execution, not a completion advantage.

Mean query time across positions is 49.4–53.7 ms for the event model, 3.2–3.5 ms for direct success, and 2.6–2.7 ms for the rule. Delays are physically propagated. The 36-episode panel uses 144,000 physical environment steps and 80.81 s wall time, with no policy training. Richer intermediate predictions incur appreciable query cost.

A separate skill comparison substitutes RB400 on 4 starts×60 candidates. Original and RB400 skills complete 34/240 and 35/240, with 13 gained and 12 lost successes; S-bend completion is 5/60 for both. This motivates skill-conditioned evaluation but does not demonstrate a robustly expanded capability domain. RB400 is not required for the main pipeline.

VIII. Discussion

The evidence supports three distinct statements. Frozen controllers complete ground-only and jump–ground tasks on the same terrain. A compact model predicts intermediate states/times and exposes errors hidden by a final label. Candidate scores are connected to actual publication and physical execution. Together these establish a working predictive planning interface, independently of whether it improves completion over a rule.

Limitations constrain stronger claims. Route templates and event order are predefined, and the map is known. Compressed history omits explicit remaining episode time. The goal label and final road factor overlap; the ranking estimate is not calibrated joint probability. One-event accuracy does not guarantee autoregressive validity. Handoff is a control switch rather than recovered stability; continuation does not explicitly estimate recovery distance. The planner performs one decision and admission recheck, not repeated model-based replanning.

Small root counts, failed right-angle tasks, and absent new-map/hardware tests limit generalization. Shared-head ablation does not replace a fixed-step dynamics comparator or removal of continuation prediction. Such

experiments are needed to attribute task-decision gains to event decomposition or recovery modeling. Current claims remain bounded to prediction and execution, with the lack of online gain explicitly reported.

IX. Conclusion

We presented an event-level predictive planner for continuous driving and gap crossing of a single-track robot. Its outputs—event occurrence, time, and post-event state—are learned from physical executions and propagated to finite candidate scores. Frozen ground/jump controllers realize those decisions, including bypass and crossing tasks with postlanding continuation. Held-out-root predictions provide quantitative model evidence, while online tests show that completion has not improved over a distance rule. The measured event errors and admission failures identify what must improve beyond demonstrating autonomous execution.

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